Rolling Class Software Setup on MAC OS

Big Data Algorithms — setup guide

These steps install the current versions of the software we use in the course on a Mac. They work on both Apple Silicon (M-series) and Intel Macs. Ubuntu users, including Ubuntu inside WSL on Windows, should follow "02_software_setup_ubuntu.qmd" instead. When you finish this guide, continue with "03_software_setup_common.qmd", which installs the R and Python packages and is the same for Ubuntu and Mac.

You should have already completed the setup in `01_linux_git_github_setup.qmd` before you begin here.

We install everything with Homebrew, the standard package manager for macOS. Run every command in the Terminal app. A few commands are too long to fit on one line, so they continue on the following lines, either after a \ or inside quotes. Copy all the lines together and the terminal runs them as one command. As of October 2026, you should end up with:

Software	Version	Check after with
R	4.6	`R --version`
Python	3.14	`python3 --version`
PostgreSQL	18	`psql --version`
Quarto	1.10	`quarto --version`
Positron	2026.09	Positron > About Positron

R

```
brew install --cask r-app
```

This downloads the official installer from CRAN, the home of R, and runs it. It asks for your Mac password. Note the `--cask`: do not use plain `brew install r`. That installs a different

build of R that cannot use CRAN's ready-made packages, so every package you install would have to be compiled on your Mac.

Check the installation:

```
R --version
```

The first line should begin with `R version 4.6`.

Now tell R where to download add-on packages from, by adding a line to `~/.Rprofile`, a file R runs every time it starts:

```
echo 'options(timeout = 600,
  repos = c(CRAN = "https://cloud.r-project.org"))' >> ~/.Rprofile
```

The `timeout` part allows each download ten minutes instead of R's default of one, which is too short for the larger packages on a slow connection.

Python

```
brew install python
python3 --version
```

This should print `Python 3.14` followed by a third number. Note that the command is `python3`, not `python`.

macOS comes with an older Python of its own, so make sure the terminal uses the one from Homebrew:

```
which python3
```

This should print `/opt/homebrew/bin/python3` (Apple Silicon) or `/usr/local/bin/python3` (Intel). If it prints `/usr/bin/python3`, open a new terminal window and try again.

Two of the Python packages we install later, `xgboost` and `lightgbm`, need a library called OpenMP, which macOS does not include. Install it now:

```
brew install libomp
```

PostgreSQL

Install PostgreSQL 18 and start the database server:

```
brew install postgresql@18
brew services start postgresql@18
```

`brew services` also restarts the server every time you log in. Homebrew does not put the PostgreSQL commands on your PATH, so add them yourself:

```
echo "export PATH=\"$(brew --prefix postgresql@18)/bin:\$PATH\"" >> ~/.zshrc
source ~/.zshrc
```

Homebrew sets up PostgreSQL so that your Mac username is the database administrator. Create a database with the same name as your username:

```
createdb $USER
```

From now on `psql` connects without any extra options:

```
psql -c "SELECT version();"
```

This should print one row beginning with `PostgreSQL 18`. Typing `psql` alone opens the interactive SQL prompt and `\q` quits it.

If `psql` cannot connect, check that the server is running with `brew services list`. The line for `postgresql@18` should say `started`.

Quarto

```
brew install --cask quarto
```

It asks for your Mac password. Check the installation:

```
quarto --version
quarto check
```

`quarto check` lists everything Quarto can find on your computer, and should show the R and Python you just installed. It may also tell you that Jupyter (for Python) and the `knitr` package (for R) are not installed yet. `knitr` is installed later, in "03_software_setup_common.qmd", and Jupyter is not needed for this course's setup.

Positron

Positron is the editor (IDE) we use for R, Python and Quarto.

```
brew install --cask positron
```

Open Positron from your Applications folder. To be able to start it from the terminal as well, press Cmd+Shift+P inside Positron to bring up the Command Palette, type "Install 'positron' command in PATH" and run that command. After you restart the terminal, `positron .` opens the current folder in Positron. Positron comes with its own copy of Quarto. We installed Quarto separately so that the `quarto` command also works in the terminal.

Continue with the common setup

Continue now with "03_software_setup_common.qmd". It installs the R and Python packages we use in the course and sets up Quarto to run R and Python in the same document. Those steps are the same on Ubuntu and on a Mac.

Updating later

Homebrew updates everything you installed with it:

```
brew update && brew upgrade
```

Positron also offers updates by itself when a new version is released.